Personal Study Analytics Dashboard Report

1 Introduction

This project analyzes personal study habits over a period of 30 days using data visualization and basic analytics techniques. The dataset contains information about study duration, focus score, study location, subjects studied, break duration, and devices used during study sessions.

The main goal of this project is to identify patterns in productivity and understand how various factors affect study performance.

2 Dataset Description

The dataset contains the following columns:

- **Date** – Date of study session
- **Start Time** – Starting time of study
- **End Time** – Ending time of study
- **Subject** – Subject studied
- **Study Hours** – Total study duration
- **Break Time Minutes** – Break duration between sessions
- **Focus Score** – Self-evaluated focus level from 1 to 10
- **Study Location** – Room, LRC, or Swadhyay
- **Device Used** – Laptop or iPad

3 Visualizations Used

The following visualizations were created:

1. Total Study Hours by Subject
2. Time vs Focus score heatmap
3. Focus Score Trend Over Time

4. Location vs Focus Score Heatmap
5. Device vs Average Focus Score
6. Study Hours Distribution by Location

4 Analysis and Observations

4.1 Subject Analysis

The bar chart of study hours showed that Mathematics and Physics consumed the highest amount of study time. Computer Science sessions were generally shorter.

4.2 Focus Trend

Higher focus scores were observed during longer study sessions.

4.3 Study Location Analysis

Among all study locations, LRC showed the highest average focus score. Swadhyay also produced good productivity levels, while Room sessions occasionally showed lower concentration levels due to distractions.

4.4 Device Analysis

Laptop-based study sessions demonstrated slightly higher focus scores compared to iPad sessions. This suggests that the laptop environment was more effective for longer and more concentrated study sessions.

5 Correlation Analysis

5.1 Study Hours vs Focus Score

A positive relationship was observed between study hours and focus score. Longer study sessions generally resulted in improved concentration and better productivity levels.

5.2 Study Location vs Productivity

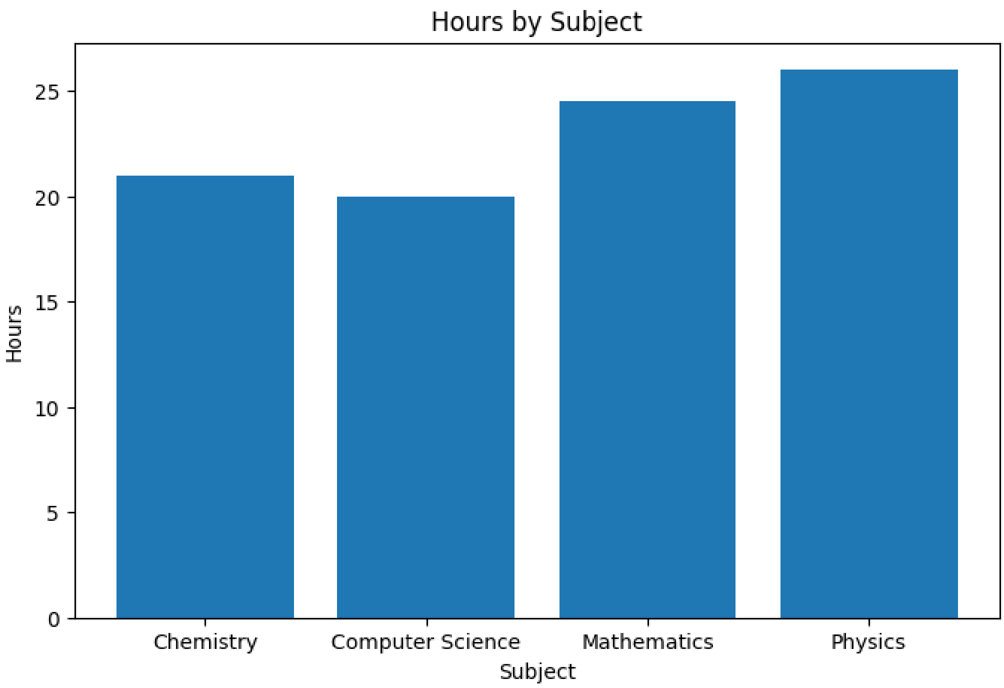
LRC and Swadhyay environments showed higher productivity levels compared to Room sessions. These locations provided fewer distractions and more consistent focus.

5.3 Device Used vs Study Efficiency

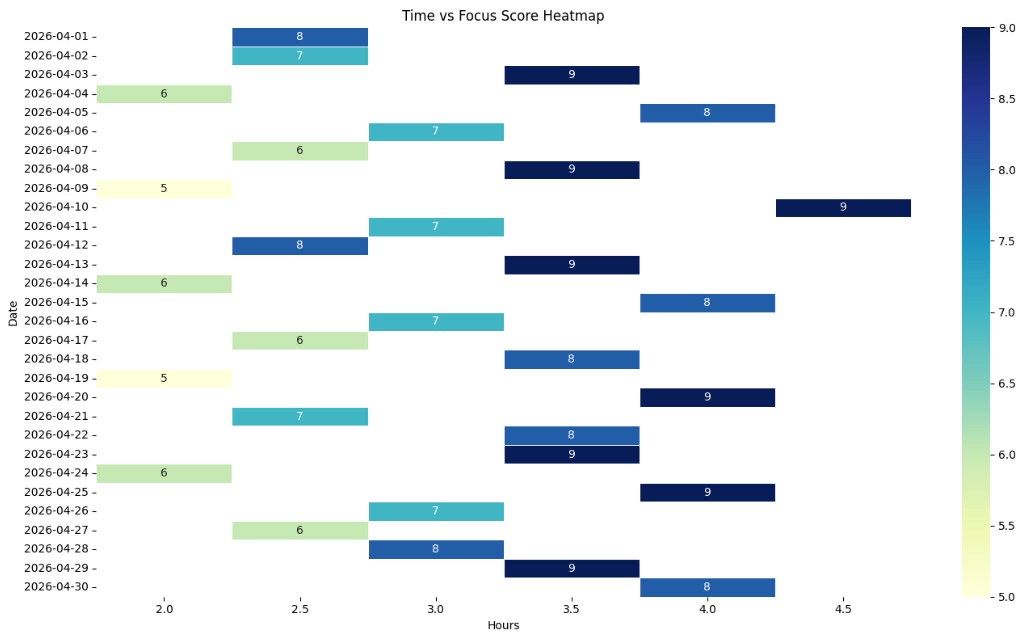
Laptop sessions demonstrated better consistency and efficiency compared to iPad sessions, especially during longer study periods.

6 plots

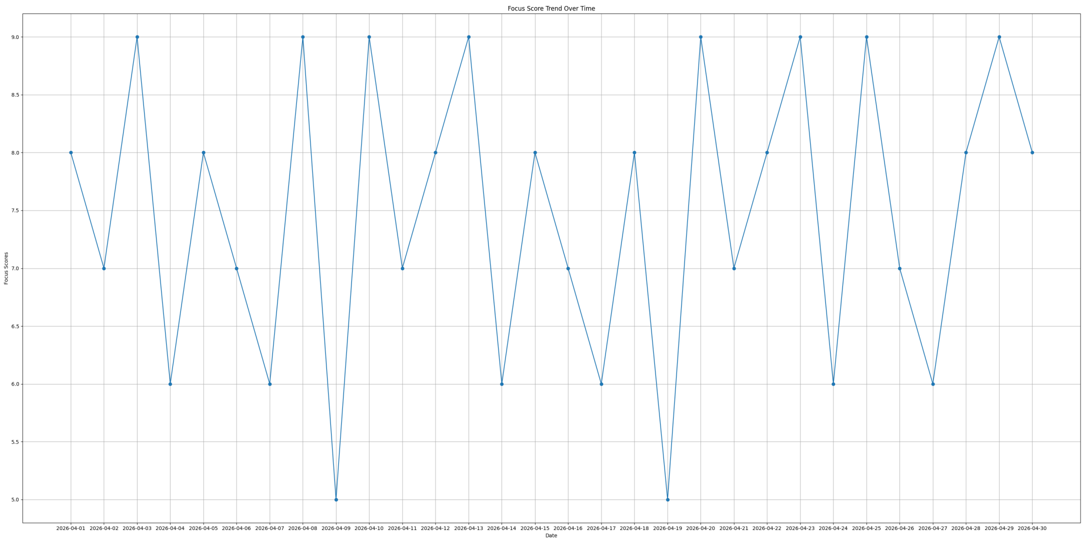
6.1 Total Study Hours by Subject



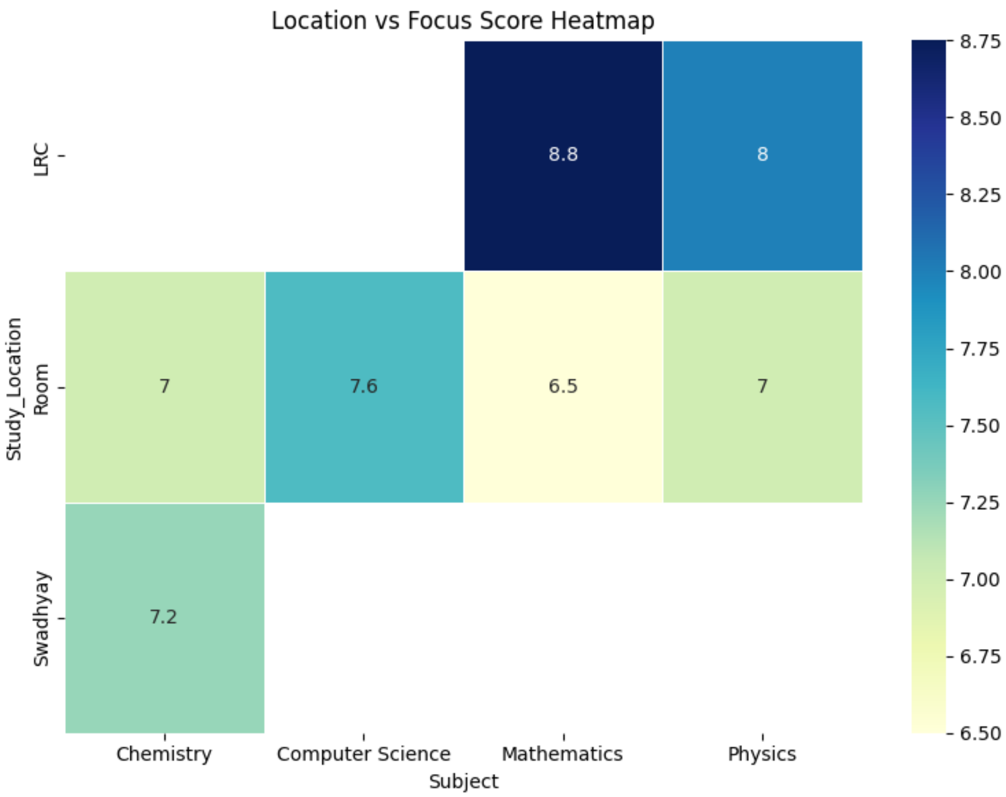
6.2 Time vs Focus score heatmap



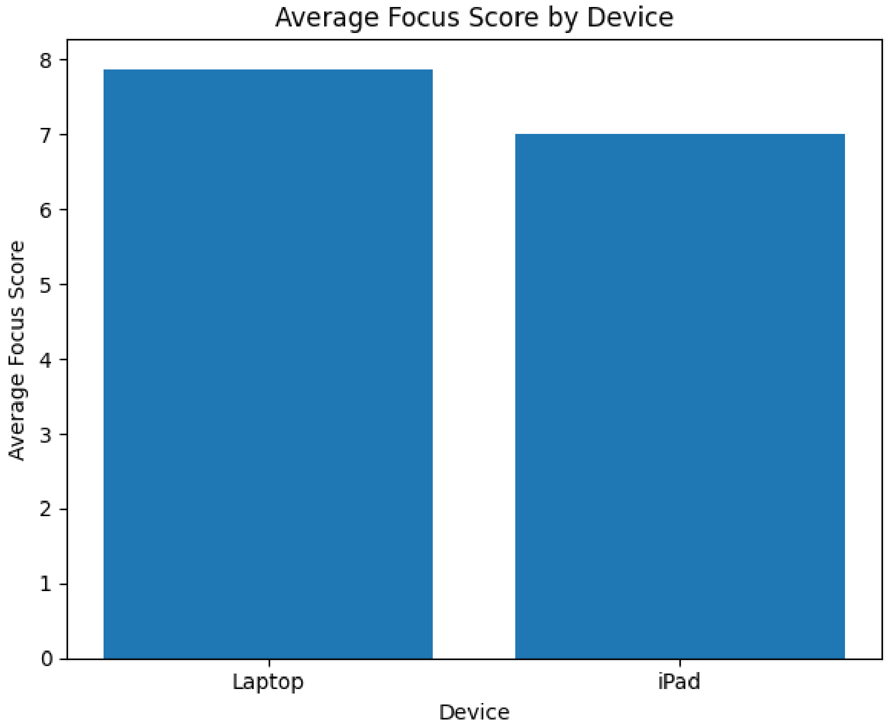
6.3 Focus Score Trend Over Time



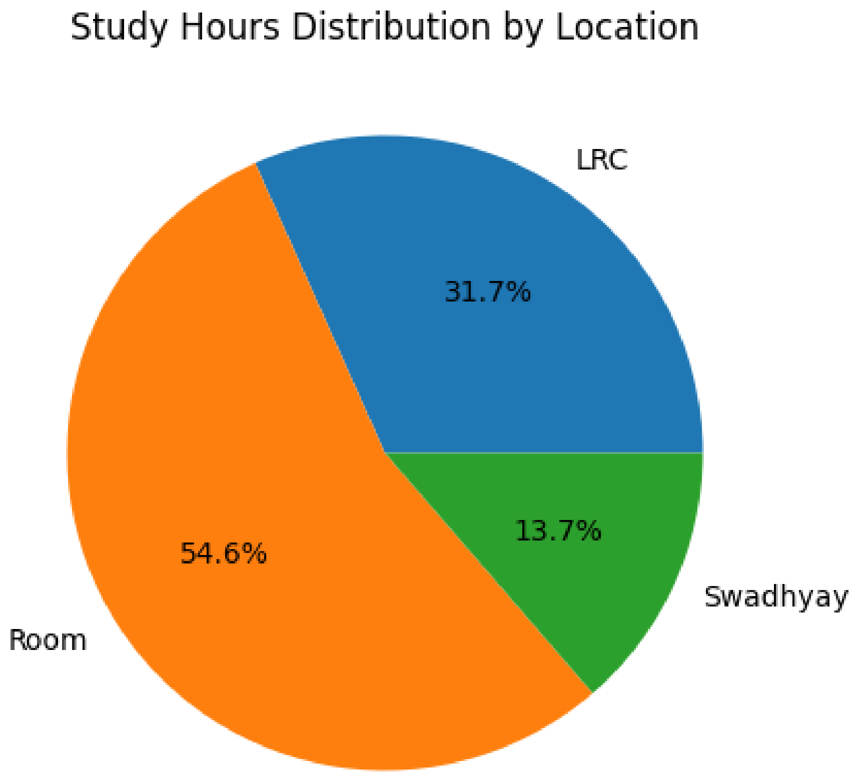
6.4 Location vs Focus Score Heatmap



6.5 Device vs Average Focus Score



6.6 Study Hours Distribution by Location



7 Conclusion

This project successfully demonstrated how data analytics can be applied to personal study habits.

The analysis showed that:

- LRC was the most productive study environment.
- Mathematics required the highest study hours.
- Laptop sessions produced better consistency and concentration.
- Moderate study sessions with controlled breaks resulted in higher productivity.

Overall, the project provided useful insights into study behavior and productivity patterns using visual analytics techniques.